

GlowBeauty Premium Skincare Audience Feed Evaluation

Findings report · Simulated customer-feed evaluation · September 2026 · All data synthetic

1 · Customer request and feed design

Objective. Reach customers likely to purchase premium skincare products while maintaining strong data quality and matchability.

Target audience. Customers with skincare engagement in the last 90 days (window: 2026-06-23 → 2026-09-21), excluding opted-out users and invalid identifiers.

Required fields

Field	Role
customer_id / email / hashed_email	Identity & matching (hashed email preferred)
zip_code, product_category, last_activity_date	Targeting & recency
engagement_score, premium_interaction_flag, total_purchases, total_spend	Prioritization

Success criteria

Criterion	Threshold	Result
Required-field completeness	≥ 95%	96.2–100% · PASS
Email/hash matchability	≥ 70%	58.8% · FAIL
Opt-out leakage	0	0 · PASS
Duplicate customer_ids in feed	0	0 · PASS

Delivery specification

Daily CSV (one row per customer, UTF-8) delivered to **s3://glowbeauty-feeds/audience/** or SFTP drop (PGP-encrypted in production), filename **glowbeauty_audience_YYYYMMDD.csv**.

2 · Feed profile

Source inputs (synthetic): 306 customer rows (300 unique + 6 duplicates), 500 events, 30 products, 150 orders.

Exclusion funnel

Stage	Customers	% of engaged
Engaged with skincare in window	166	100%
Excluded — opted out	16	9.6%
Excluded — missing/invalid email	19	11.4%
Final feed	131	78.9%

21.1% of engaged customers are excluded before delivery — every exclusion is a deliberate quality gate, not data loss.

Audience composition by engagement tier

Tier (score)	Customers	% of feed	Avg score	Premium interactors
High (≥ 10)	20	15.3%	11.7	8
Medium (4–9)	29	22.1%	5.4	18
Low (< 4)	82	62.6%	1.6	33

Required-field completeness

Field	Completeness	Threshold	Status
zip_code	96.2%	≥ 95%	PASS
product_category	100%	≥ 95%	PASS
last_activity_date	100%	≥ 95%	PASS
engagement_score	100%	≥ 95%	PASS

Identifier coverage (feed, n=131): valid email 100% (by construction) · hashed email **58.8%** (77/131).

3 · Key data-quality findings

- **41.2% of feed rows lack a hashed email** (54 of 131). The single largest drag on matchability — these customers are reachable by email but not by hashed matching.
- **12.4% of the customer base has unusable emails:** 7.7% missing, 4.7% invalid format (missing @, double @, trailing dot). 19 engaged customers were excluded for this reason.
- **6 duplicate customer_id pairs** in the source were resolved with a survivorship rule (usable email → hashed email → most recent record); 0 duplicates remain in the feed. Without dedupe, audience counts would be inflated ~2%.
- **3.8% of feed rows are missing ZIP codes** (5 of 131) — within the completeness bar, but a targeting gap for geo-based suppression/expansion.
- **16 opted-out customers had in-window skincare engagement** and were correctly excluded; opt-out leakage in the feed is 0.
- **35 events were unusable for audience building:** 4 future-dated, 31 outdated (>90d); 20 events and 4 orders referenced missing/unknown customers, 14 events and 4 orders referenced unknown products — all dropped by inner joins, none silently included.

4 · Gap analysis — expected vs. actual

Measure	Expected (target)	Actual	Verdict
Required-field completeness	≥ 95%	96.2–100%	PASS
Email/hash matchability	≥ 70%	58.8% (77/131)	FAIL
Opt-out leakage	0	0	PASS

Diagnosis

- The feed **meets completeness requirements but misses the 70% matchability target** because email and hashed-email coverage is insufficient for part of the target population.
- Root cause is **upstream, not in the audience logic**: 54 feed rows were never provisioned a hashed email at the source. The feed query keeps these rows deliberately (reach over match rate) and flags them — see **sql/03** section E and **python/feed_validation.py**.
- Secondary contributors: 19 engaged customers excluded for missing/invalid emails (11.4%), and 6 duplicate source records that would have inflated counts without the survivorship rule.
- What would *not* fix it: widening the engagement window or lowering the score bar would add rows, but rows without identifiers still cannot match — matchability is an **identifier-supply** problem, not an audience-size problem.

5 · Recommendations and tradeoffs

- **Require hashed email for match-guaranteed eligibility** and backfill hashes for the 54 affected customers (hash is derivable from the valid emails on file).
- **Add email-format and identifier validation upstream** at ingestion — reject or quarantine invalid formats before they reach audience building (currently 4.7% of the base).
- **Keep the dedupe survivorship rule** and monitor duplicate rate; alert if it exceeds 1%.
- **Offer two audience options** so the customer chooses the tradeoff explicitly:

Option	Size (of 131)	Matchability	Best for
A · High-matchability	77	~100% of rows	Lookalike seeding, suppression, identity resolution
B · Expanded reach	131	58.8%	Broad prospecting where reach outweighs match precision

The `matchable_flag` column in `output/glowbeauty_audience_feed_clean.csv` implements this split without rebuilding the feed.

- **Monitor daily delivery health:** record volumes, null rates, and identifier coverage — see the monitoring plan on the next page.

6 · Production monitoring plan

Daily checks on every feed delivery; alert on breach, page on critical:

Monitor	Metric	Threshold / alert
Record-count variance	Rows vs. 7-day rolling median	Alert beyond $\pm 15\%$; page beyond $\pm 30\%$
Required-field completeness	% populated per required field	Alert < 95% on any field
Duplicate rate	% duplicate customer_ids	Alert > 0% (feed must be unique)
Email / hash coverage	% valid email; % hashed email	Alert if hash coverage drops > 2 pts vs. baseline
Opt-out leakage	Opted-out ids present in feed	Page on any count > 0
Delivery timeliness	File landed by 06:00 ET	Alert at 06:15; page at 07:00
Schema-change alerts	Column set / types vs. contract	Page on any undeclared change

Runbook

- Quarantine (do not deliver) any file failing schema, duplicate, or opt-out checks.
- Completeness or coverage breaches deliver with a **quality advisory header** and a ticket to the source team.
- Weekly: review exclusion-funnel drift (opt-out / bad-identifier rates) and tier mix for audience decay.

Reproduce everything: `data/generate_synthetic_data.py` → `sql/01-03` → `python/feed_validation.py`. Full instructions in README.md.